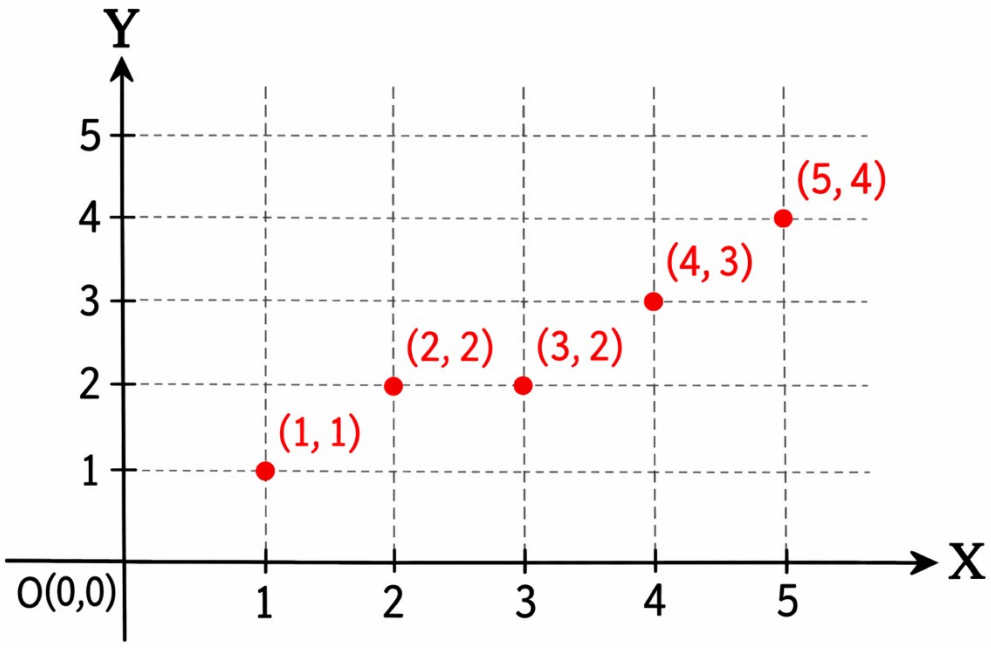
Unit 3 - Session 3

SLO 1: Describe least-squares line fitting and its limitations.

Duration: 10 Minutes Activity

Question:



Q1. Do the given five points lie exactly on a single straight line?

Answer: No. The points are not exactly collinear.

Q2. What is the objective of least-squares line fitting?

Answer: To find the best-fit line by minimizing the sum of squared errors.

Q3. What type of error is minimized in the least-squares method?

Answer: Squared residual error.

Q4. Why does the least-squares fitted line not pass through all the points?

Answer: Because the points are not all on one straight line.

Q5. Which points contribute most to the fitting error?

Answer: Point (3, 2), because it has the largest residual.

Q6. What happens to the best-fit line when an outlier point is added far from the main cluster?

Answer: The best-fit line shifts toward the outlier.

Q7. Why is least-squares line fitting considered sensitive to outliers?

Answer: Squaring residuals gives large errors more weight.

Q8. State one limitation of least-squares line fitting in computer vision applications.

Answer: It is sensitive to outliers.